



# MATHEMATICS

2014 HSC Course Assessment Task 2

Monday March 10, 2014

## General instructions

- Working time – 55 minutes.  
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets + answer sheet used in the correct order within this paper and hand to examination supervisors.

## SECTION I

- Mark your answers on the answer sheet provided (numbered as page 5)

## SECTION II

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

**STUDENT NUMBER:** ..... **# BOOKLETS USED:** .....

**Class** (please ✓)

☐ 12M2A – Mr Lin

☐ 12M3A – Mr Zuber

☐ 12M4A – Ms Ziariaris

☐ 12M3B – Mr Berry

☐ 12M4B – Mr Lam

☐ 12M2B – Mr Weiss

☐ 12M3C – Mr Lowe

☐ 12M4C – Mr Ireland

Marker's use only.

| QUESTION | 1-5            | 6              | 7              | 8              | 9               | 10             | Total           | % |
|----------|----------------|----------------|----------------|----------------|-----------------|----------------|-----------------|---|
| MARKS    | $\overline{5}$ | $\overline{9}$ | $\overline{8}$ | $\overline{9}$ | $\overline{10}$ | $\overline{8}$ | $\overline{44}$ |   |

## Section I: Objective response

Mark your answers on the multiple choice sheet provided.

**Marks**

1. What is the value of  $e^{-0.2}$  correct to 3 decimal places?

**1**

(A) 0.818                      (B) 0.819                      (C) 1.221                      (D) 1.222

2. Which expression is equivalent to  $\int \sqrt{5x+1} \, dx$ ?

**1**

(A)  $\frac{1}{10}(5x+1)^{\frac{3}{2}} + C$                       (C)  $\frac{3}{10}(5x+1)^{\frac{3}{2}} + C$

(B)  $\frac{2}{15}(5x+1)^{\frac{3}{2}} + C$                       (D)  $\frac{2}{3}(5x+1)^{\frac{3}{2}} + C$

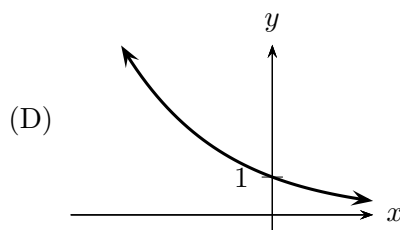
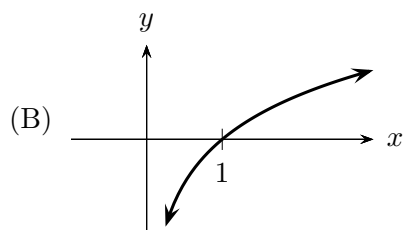
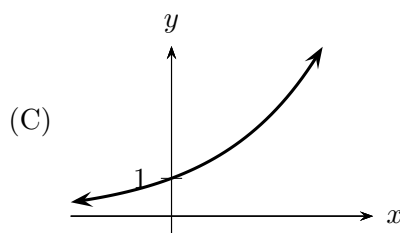
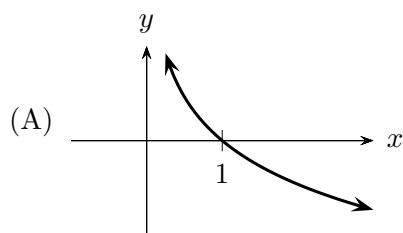
3. If  $f(x) = e^{-x} - 2e^{-2x}$ , what is the value of  $f(\ln 2)$ ?

**1**

(A) 0                      (B) 4                      (C) 6                      (D) 10

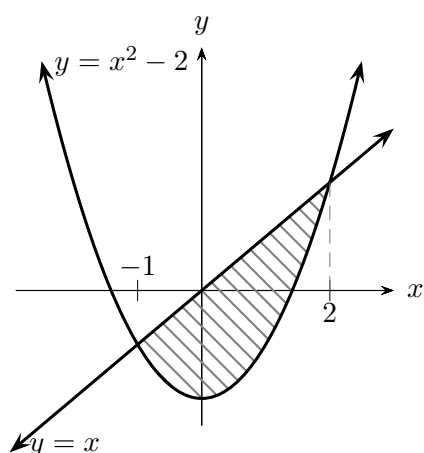
4. Which of the following best represents the shape of the graph  $y = \log_e x$ ?

**1**



5. Which of the following integrals is representative of the shaded area?

**1**



(A)  $\int_{-1}^2 (x - x^2 + 2) \, dx$

(B)  $\int_{-1}^2 (x^2 + x - 2) \, dx$

(C)  $\int_{-1}^2 (x^2 - x - 2) \, dx$

(D)  $\int_{-1}^2 (x - x^2 - 2) \, dx$

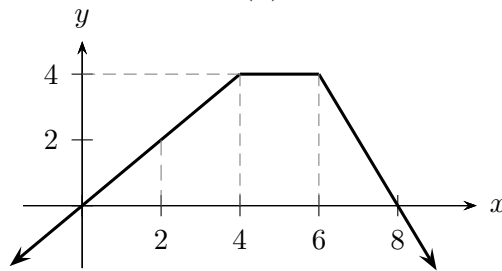
**End of Section I.**  
**Examination continues overleaf.**

**Question 6** (9 Marks)

Commence a NEW page.

**Marks**

- (a) The function
- $y = f(x)$
- is shown in the following diagram.



Evaluate:

i.  $\int_0^8 f(x) dx.$

**2**

ii.  $\int_{-2}^8 f(x) dx.$

**1**

- (b) Evaluate
- $\int_1^4 (3x - 2)^4 dx.$

**3**

- (c) Find the area enclosed by the curves
- $y = x^2 - x$
- and
- $y = 5x - x^2$
- .

**3****Question 7** (8 Marks)

Commence a NEW page.

**Marks**

- (a) Find the primitives of the following:

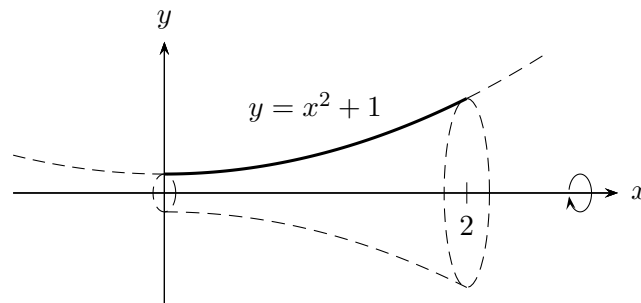
i.  $\frac{x^4 - 6}{x^3}.$

**2**

ii.  $\sqrt{x}.$

**2**

- (b) Find the volume of the solid of revolution generated when the curve
- $y = x^2 + 1$
- is rotated about the
- $x$
- axis between
- $x = 0$
- and
- $x = 2$
- .

**4****Question 8** (9 Marks)

Commence a NEW page.

**Marks**

- (a) Given the parabola
- $x^2 = -6y$
- , find:

i. the coordinates of the vertex.

**1**

ii. the coordinates of the focus.

**1**

iii. the equation of the directrix.

**1**

- (b) Find the coordinates of the vertex and focus of the parabola

**3**

$$x^2 - 6x - 3y - 12 = 0$$

- (c) A parabola has its focus at
- $(1, -4)$
- and the directrix is the line
- $y = 6$
- .

**3**

Find the equation of the parabola.

- Question 9** (10 Marks) Commence a NEW page. **Marks**
- (a) Evaluate the following, giving your answer in simplest form.
- $\frac{d}{dx} \left( \frac{x}{e^x} \right).$  **2**
  - $\frac{d}{dx} (x^2 e^{-(x+2)}).$  **2**
  - $\frac{d}{dx} \left( \log_e \left( \frac{3x+4}{2x-1} \right) \right).$  **2**
- (b) For the function  $y = \log_e(2x+1)$ ,
- Find the domain of the function. **1**
  - Find the range of the function. **1**
  - Sketch the function, showing all important information. **2**

- Question 10** (8 Marks) Commence a NEW page. **Marks**
- (a) Evaluate:
- $\int_0^2 3e^{1-5x} dx.$  **2**
  - $\int xe^{x^2} dx.$  **2**
- (b) Given  $y = \log_e x$ ,
- |            |   |   |   |   |   |
|------------|---|---|---|---|---|
| $x$        | 1 | 2 | 3 | 4 | 5 |
| $\log_e x$ |   |   |   |   |   |
- Copy and complete the table above, giving the values correct to 3 decimal places. **1**
  - Use Simpson's Rule with five function values to estimate **3**

$$\int_1^5 \log_e x dx$$

**End of paper.**

## Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g “●”

**STUDENT NUMBER:** .....

**Class** (please ✓)

☐ 12M2A – Mr Lin

☐ 12M3A – Mr Zuber

☐ 12M4A – Ms Ziazaris

☐ 12M3B – Mr Berry

☐ 12M4B – Mr Lam

☐ 12M2B – Mr Weiss

☐ 12M3C – Mr Lowe

☐ 12M4C – Mr Ireland

**1** – ☐ A ☐ B ☐ C ☐ D

**2** – ☐ A ☐ B ☐ C ☐ D

**3** – ☐ A ☐ B ☐ C ☐ D

**4** – ☐ A ☐ B ☐ C ☐ D

**5** – ☐ A ☐ B ☐ C ☐ D

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right) + C$$

NOTE:  $\ln x = \log_e x$ ,  $x > 0$

## Suggested Solutions

### Section I

1. (D) 2. (B) 3. (D) 4. (C) 5. (A)

### Section II

**Question 5** (Berry)

(a)

# Marking

Q1 B

2 B

3 A

4 B

5 A

Q6 Lin

Q7 Lin

Q8 Weiss

Q9 Zuber

Q10 Ziaziaris

Q6) a) i)  $2 + 6 + 8 + 4$

$$= 20$$

2

ii)  $20 - 2$

$$= 18$$

1

b)  $\int_1^4 (3x-2)^4 dx$

$$= \left[ \frac{(3x-2)^5}{15} \right]_1^4$$

1

$$= \frac{10^5}{15} - \frac{1}{15}$$

1

$$= \frac{10^5 - 1}{15}$$

1

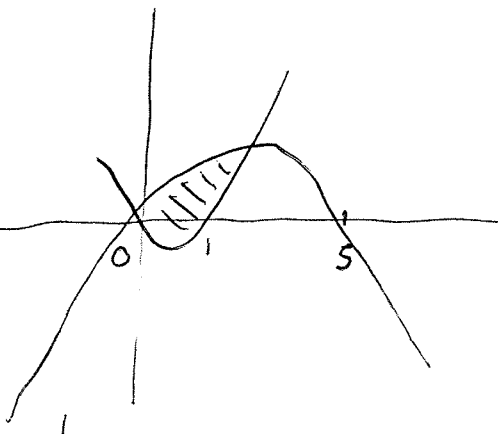
c)

$$5x - x^2 = x^2 - x$$

$$2x^2 - 6x = 0$$

$$2x(x-3) = 0$$

$$\therefore x = 0 \text{ or } 3$$



$$\int_0^3 (5x - x^2 - (x^2 - x)) dx$$

$$\int_0^3 (6x - 2x^2) dx$$

$$= \left[ 3x^2 - \frac{2x^3}{3} \right]_0^3$$

$$= 27 - 18 = 9 \text{ u}^2$$

$$Q7) a) i) \int x - 6x^{-3} dx$$

$$= \frac{x^2}{2} + 3x^{-2} + C$$

$$ii) \int x^{\frac{1}{2}} dx$$

$$= \frac{2x^{\frac{3}{2}}}{\frac{3}{2}} + C$$

$$b) V = \pi \int_0^2 y^2 dx$$

$$= \pi \int_0^2 x^4 + 2x^2 + 1 dx$$

$$= \pi \left[ \frac{x^5}{5} + \frac{2x^3}{3} + x \right]_0^2$$

$$= \left( \frac{32}{5} + \frac{16}{3} + 2 \right) \pi u^3$$

$$= 13 \frac{11}{15} \pi u^3$$

Q8)

a) i)  $x^2 = 4ay$

$$x^2 = -6y$$

$$a = -\frac{3}{2}$$

i)  $V(0,0)$

ii)  $S(0, -\frac{3}{2})$

iii)  $y = \frac{3}{2}$

b)  $x^2 - 6x - 3y - 12 = 0$

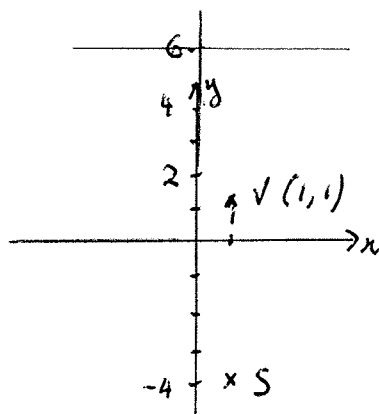
$$x^2 - 6x + 9 = \cancel{x^2} 3y + 12 + 9$$

$$(x-3)^2 = 3(y+7)$$

$\therefore$  Vertex  $(3, -7)$

Focus  $(3, -6\frac{1}{4})$

c)



$$(x-1)^2 = -20(y-1)$$

Q9)

a)

i/

$\frac{d}{dx}$

$\frac{x}{e^x}$

$$= \frac{e^x \cdot 1 - x e^x}{e^{2x}}$$

$$= \frac{x^x (1-x)}{e^{2x}}$$

$$= \frac{1-x}{e^x}$$

ii/

$\frac{d}{dx}$

$x^2 e^{-(x+2)}$

$$= 2x e^{-(x+2)} + x^2 e^{-(x+2)}$$

$$= x e^{-(x+2)} (2-x)$$

$$\text{iii/ } \frac{d}{dx} \left( \ln(3x+4) - \ln(2x-1) \right)$$

$$= \frac{3}{3x+4} - \frac{2}{2x-1}$$

$$= \frac{6x-3-6x-8}{(3x+4)(2x-1)}$$

$$= \frac{-11}{(3x+4)(2x-1)}$$

Q10)

a) i/  $\int_0^2 3e^{1-5x} dx$

$$= \frac{-3}{5} e^{1-5x} \Big|_0^2$$

$$= \frac{-3}{5} e^9 + \frac{3}{5} e^1$$

ii/  $\int x e^{x^2} dx$

$$= \frac{1}{2} \int 2x e^{x^2} dx$$

$$= \frac{1}{2} e^{x^2} + C$$

b) i/

| $x$     | 1 | 2    | 3     | 4     | 5     |
|---------|---|------|-------|-------|-------|
| $\ln x$ | 0 | .693 | 1.099 | 1.386 | 1.609 |

ii/  $\int_1^5 \log_e x dx = \frac{1}{3} \left[ 0 + 1.609 + 4(.693 + 1.386) + 2(1.099) \right]$

$$= 4.041$$